THE REQUIREMENTS FOR ACQUIRING THE PRODUCT(S) DESCRIBED HEREIN SHALL CONSIST OF THIS SPECIFICATION SHEET AND THE ISSUE OF THE FOLLOWING SPECIFICATION LISTED IN THAT ISSUE OF THE DODISS SPECIFIED IN THE SOLICITATION: FF-S-86

... IS APPROVED FOR USE BY ALL DEPARTMENTS AND AGENCIES OF THE ... ARTMENT OF DEFENSE.

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OMB No. 0704-0188

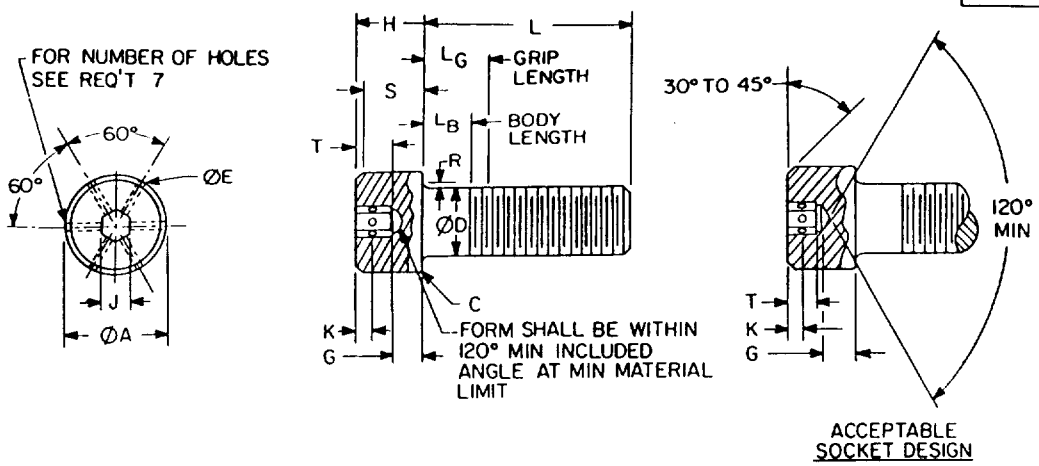


TABLE I. DASH NUMBERS AND DIMENSIONS

NOMINAL SIZE			#4 (.112)	#6 (.138)	#8 (.164)	#10 (.190)
THREAD SIZE			.112-40UNC-3A	.138-32UNC-3A	.164-32UNC-3A	.190-24UNC-3A
ØD	BODY	MAX	.1120	.1380	.1640	.1900
		MIN	.1075	.1329	.1585	.1840
ØA	HEAD	MAX	.183	.226	.270	.312
		MIN	.176	.218	.262	.303
H	HEAD HEIGHT	MAX	.112	.138	.164	.190
		MIN	.108	.134	.159	.185
S	HEAD SIDE HEIGHT	MIN	.101	.124	.148	.171
J	SOCKET WIDTH ACROSS FLATS	MAX	.0952	.1111	.1426	.1587
		MIN	.0937	.1094	.1406	.1562
T	KEY ENGAGEMENT	MIN	.051	.064	.077	.090
G	WALL THICKNESS	MIN	.038	.047	.056	.065
C	CHAMFER OR RADIUS	MAX	.005	.005	.005	.005
R	FILLET EXTENSION	MAX	.009	.010	.012	.014
		MIN	.005	.006	.007	.009
R	HOLE LOCATION	MAX	.040	.050	.060	.065
		MIN	.026	.035	.040	.045
ØE	HOLE	MAX	.039	.039	.050	.050
		MIN	.033	.033	.044	.044
	RECOMMENDED WIRE DIAMETER	MAX	.028	.028	.040	.040
		MIN	.020	.020	.032	.032
	ALIGNMENT PLUG DIAMETER 1/		.025	.025	.030	.030
	TENSILE STRENGTH, LBS 2/	MIN	1,090	1,640	2,520	3,150

L LENGTH	TOLERANCE	#4 DASH NO	#4 LG NAL.	#4 LB MIN	#6 DASH NO	#6 LG MAX	#6 LB MIN	#8 DASH NO	#8 LG MAX	#8 LB MIN	#10 DASH NO	#10 LG MAX	#10 LB MIN
.250		1			6			13					
.375		2			7			14			22		
.500	+0.00	3			8			15			23		
.625	-0.03	4			9			16			24		
.750		5			10			17			25		
.875					11			18			26		
1.000					12			19			27		
1.250								20	.38	.22	28	.38	.17
1.500	+0.00							21	.38	.22	29	.38	.17
1.750	-0.04										30	.88	.67
2.000											31	.88	.67

1/ SEE REQUIREMENT 6

2/ SEE REQUIREMENT 3

Ⓓ ENTIRE STANDARD REVISED

INCH - POUND

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- NAVY- OS
AIR FORCE- 99 DLA-
REVIEW: MC,MI,SH,IS,11
USER: AS,AT,ME
PROJECT NUMBER: 5305-1916

MILITARY SPECIFICATION SHEET

TITLE

SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNC-3A)

SPECIFICATION SHEET NUMBER
MS24677D 2 JAN 92

SUPERSEDING MS24677C 5 MARCH 1986 AND MS24676

AMSC- N/A FSC - 5305

DISTRIBUTION STATEMENT
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TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			1/4 (.250)	5/16 (.3125)	3/8 (.375)	7/16 (.4375) 3/
THREAD SIZE			.250-20UNC-3A	.3125-18UNC-3A	.375-16UNC-3A	.4375-14UNC-3A
ØD	BODY	MAX	.2500	.3125	.3750	.4375
		MIN	.2435	.3053	.3678	.4294
ØA	HEAD	MAX	.375	.469	.562	.656
		MIN	.365	.457	.550	.642
H	HEAD HEIGHT	MAX	.250	.312	.375	.438
		MIN	.244	.306	.368	.430
S	HEAD SIDE HEIGHT	MIN	.225	.281	.337	.394
J	SOCKET WIDTH ACROSS FLATS	MAX	.1900	.2530	.3160	.3790
		MIN	.1875	.2500	.3125	.3750
T	KEY ENGAGEMENT	MIN	.120	.151	.182	.213
G	WALL THICKNESS	MIN	.095	.119	.143	.166
C	CHAMFER OR RADIUS	MAX	.008	.008	.008	.010
R	FILLET EXTENSION	MAX	.014	.017	.020	.023
		MIN	.009	.012	.015	.018
K	HOLE LOCATION	MAX	.085	.104	.123	.141
		MIN	.065	.084	.103	.121
ØE	HOLE	MAX	.050	.050	.067	.067
		MIN	.044	.044	.061	.061
	RECOMMENDED WIRE DIAMETER	MAX	.040	.040	.057	.057
		MIN	.032	.032	.045	.045
	ALIGNMENT PLUG DIAMETER 1/		.030	.030	.051	.051
	TENSILE STRENGTH, LBS 2/	MIN	5,400	8,900	13,200	18,100

L LENGTH	TOLERANCE SIZE .375 OR UNDER	TOLERANCE SIZE .4375 AND OVER	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN
.375			32			46								
.500			33			47								
.625	+0.00	+0.00	34			48						80		
.750	-0.03	-0.03	35			49						81		
.875			36			50						82		
1.000			37			51						83		
1.250			38			52						84		
1.500			39	.50	.25	53						85		
1.750	+0.00	+0.00	40	.50	.25	54	.62	.35	69	.50	.19	86		
2.000	-0.04	-0.06	41	1.00	.75	55	.62	.35	70	.50	.19	87		
2.250			42	1.00	.75	56	1.12	.85	71	1.00	.69	88	.62	.27
2.500			43	1.50	1.25	57	1.12	.85	72	1.00	.69	89	.62	.27
2.750			44	1.50	1.25	58	1.62	1.35	73	1.50	1.19	90	1.12	.77
3.000			45	2.00	1.75	59	1.62	1.35	74	1.50	1.19	91	1.12	.77
3.250	+0.00	+0.00				60	2.12	1.85	75	2.00	1.69	92	1.62	1.27
3.500	-0.06	-0.08				61	2.12	1.85	76	2.00	1.69	93	1.62	1.27
4.000									77	2.50	2.19	94	2.12	1.77
4.500									78	3.00	2.69	95	2.62	2.27
5.000									79	3.50	3.19			

1/ SEE REQUIREMENT 8

2/ SEE REQUIREMENT 9

3/ INACTIVE FOR NEW DESIGN

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- NAVY- OS
AIR FORCE-99 DLA-
REVIEW: MC,MI,SH,IS,11
USER: AS,AT,ME
PROJECT NUMBER: 5305-1916
DISTRIBUTION STATEMENT
A. Approved for public release; distribution is unlimited.

MILITARY SPECIFICATION SHEET

TITLE
SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNC-3A)

SPECIFICATION SHEET NUMBER
MS24677D 2 JAN 92
SUPERSEDING MS24677C 5 MARCH 1986 AND MS24676
AMSC- N/A FSC - 5305

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DEPARTMENT OF DEFENSE

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Form Approved
OMB No. 0704-0188

TABLE I. DASH NUMBERS AND DIMENSIONS - CONTINUED

NOMINAL SIZE			1/2 (.500)	5/8 (.625)	3/4 (.750)	7/8 (.875) 3/	1 (1.000)
THREAD SIZE			.500-13UNC-3A	.625-11UNC-3A	.750-10UNC-3A	.875-9UNC-3A	1.000-8UNC-3A
ØD	BODY	MAX	.5000	.6250	.7500	.8750	1.0000
		MIN	.4919	.6163	.7406	.8647	.9886
ØA	HEAD	MAX	.750	.938	1.125	1.312	1.500
		MIN	.735	.921	1.107	1.293	1.479
H	HEAD HEIGHT	MAX	.500	.625	.750	.875	1.000
		MIN	.492	.616	.740	.864	.988
S	HEAD SIDE HEIGHT	MIN	.450	.562	.675	.787	.900
J	SOCKET WIDTH ACROSS FLATS	MAX	.3790	.5050	.6310	.7570	.7570
		MIN	.3750	.5000	.6250	.7500	.7500
T	KEY ENGAGEMENT	MIN	.245	.307	.370	.432	.495
G	WALL THICKNESS	MIN	.190	.238	.285	.333	.380
C	CHAMFER OR RADIUS	MAX	.010	.010	.010	.015	.015
R	FILLET EXTENSION	MAX	.026	.032	.039	.044	.050
		MIN	.020	.024	.030	.034	.040
K	HOLE LOCATION	MAX	.160	.198	.235	.273	.310
		MIN	.140	.178	.215	.253	.290
ØE	HOLE	MAX	.067	.067	.097	.097	.097
		MIN	.061	.061	.091	.091	.091
	RECOMMENDED WIRE DIAMETER	MAX	.057	.057	.080	.080	.080
		MIN	.045	.045	.064	.064	.064
	ALIGNMENT PLUG DIAMETER 1/		.051	.051	.081	.081	.081
	TENSILE STRENGTH, LBS 2/	MIN	24,100	38,400	56,800	78,500	103,000

L LENGTH	TOLERANCE SIZE .750	TOLERANCE SIZE .875	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN	DASH NO	L_G MAX	L_B MIN
	OR UNDER	AND OVER															
.500			96														
.625	+0.00	+0.00	97														
.750	-0.03	-0.05	98														
.875			99														
1.000			100			116											
1.250			101			117			132								
1.500			102			118			133								
1.750	+0.00	+0.00	103			119			134								
2.000	-0.06	-0.10	104			120			135			147			159		
2.250			105	.75	.36	121			136			148			160		
2.500			106	.75	.36	122	.75	.30	137			149			161		
2.750			107	.75	.36	123	.75	.30	138			150			162		
3.000			108	1.50	1.12	124	.75	.30	139	1.00	.50	151			163		
3.250			109	1.50	1.12	125	1.50	1.04	140	1.00	.50	152	1.00	.44	164		
3.500			110	1.50	1.12	126	1.50	1.04	141	1.00	.50	153	1.00	.44	165	1.00	.38
4.000	+0.00	+0.00	111	2.25	1.86	127	2.25	1.80	142	2.00	1.50	154	1.00	.44	166	1.00	.38
4.500	-0.08	-0.14	112	3.00	2.62	128	2.25	1.80	143	2.00	1.50	155	2.00	1.44	167	2.00	1.38
5.000			113	3.00	2.62	129	3.00	2.54	144	3.00	2.50	156	2.00	1.44	168	2.00	1.38
5.500			114	3.75	3.36	130	3.75	3.30	145	3.00	2.50	157	3.00	2.44	169	3.00	2.38
6.000			115	4.50	4.12	131	3.75	3.30	146	4.00	3.50	158	3.00	2.44	170	3.00	2.38

1/ SEE REQUIREMENT 8

2/ SEE REQUIREMENT 9

3/ INACTIVE FOR NEW DESIGN

INTERCHANGEABILITY DATA

EFFECTIVE 7 JULY 1975, MS24676 WAS CANCELLED AND SUPERSEDED BY MS24677

MS24676 AND THE SCREWS COVERED BY THE PART NUMBERS LISTED IN THE STANDARD ARE CANCELLED AFTER THE DATE INDICATED ON THIS DOCUMENT.

USE ONLY THE PART NUMBER FROM MS24677 FOR DESIGN AND REPLACEMENT.

CANCELLED MS24676 SCREWS CANNOT ALWAYS REPLACE MS24677 SCREWS AND SHOULD BE USED ONLY WHERE APPLICABLE UNTIL EXISTING STOCK IS DEPLETED.

MS24677 SCREWS SHALL BE USED IN LIEU OF THE CANCELLED MS24676 SCREWS OF THE SAME DASH NUMBERS. FOR EXAMPLE:

SCREWS MS24676-1 THROUGH MS24676-170
ARE CANCELLED AND SUPERSEDED BY
SCREWS MS24677-1 THROUGH MS24677-170

PREPARING ACTIVITY: ARMY-AR
CUSTODIANS: ARMY- NAVY- OS
AIR FORCE- 99 DLA-
REVIEW: MC,MI,SH,IS,11
USER: AS,AT,ME
PROJECT NUMBER: 5305-1916

MILITARY SPECIFICATION SHEET

TITLE
SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNC-3A)

SPECIFICATION SHEET NUMBER
MS24677D 2 JAN 92
SUPERSEDING MS24677C 5 MARCH 1986 AND MS24676
AMSC- N/A FSC - 5305

DISTRIBUTION STATEMENT
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Form Approved
OMB No. 0704-0188

REQUIREMENTS:

1. MATERIAL: ALLOY STEEL IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

2. PROTECTIVE COATING AND SURFACE TREATMENT: UNCOATED, CADMIUM PLATE IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 3, OR ALUMINUM ION VAPOR DEPOSITED IN ACCORDANCE WITH MIL-C-83488, TYPE II, CLASS 3 (SEE REQT 11).

   (a) DO NOT USE CADMIUM PLATED SCREWS FOR TEMPERATURES ABOVE 450°.

   (b) ALUMINUM ION VAPOR DEPOSITED COATING CAN BE USED FOR TEMPERATURES UP TO 850°F.

3. HARDNESS: IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

4. THREADS: THREADS SHALL BE IN ACCORDANCE WITH FED-STD-H28/2. ACCEPTABILITY OF SCREW THREADS SHALL BE IN IN ACCORDANCE WITH FED-STD-H28/20, SYSTEM 22.

5. THREAD LENGTH: SCREWS ABOVE HEAVY LINE SHALL HAVE COMPLETE (FULL FORM) THREADS EXTENDING WITHIN TWO THREADS OF THE HEAD.

6. HEAD: HEAD MAY OR MAY NOT BE KNURLED AT MANUFACTURER'S OPTION. TYPE OF KNURLING OPTIONAL. DIAMETER OF KNURLED HEAD SHALL NOT EXCEED THE MAXIMUM HEAD DIAMETER SPECIFIED.

7. DRILLED HOLE DATA:

   (a) SCREWS #6 (.138) SHALL HAVE ONE (1) HOLE DRILLED THROUGH THE HEAD. SCREWS #8 (.164) AND LARGER SHALL HAVE (3) HOLES DRILLED THROUGH THE HEAD.

   (b) ON SCREWS SMALLER THAN #8 (.164) THE DRILLED HOLE SHALL BE CENTERED AS CLOSE AS PRACTICABLE ON THE FLAT OF THE HEXAGON SOCKET. ON SCREWS #8 (.164) AND LARGER THE DRILLED HOLES SHALL BE WITHIN THE FLATS ON THE HEXAGON SOCKET AND SHALL NOT BREAK THE CORNERS OF THE HEXAGON SOCKET.

   (c) EDGES OF THE HOLES ON THE OUTSIDE OF THE HEAD SHALL BE CHAMFERED 45° BY 0.010 TO 0.020 DEEP.

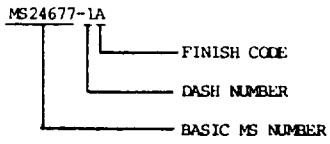
8. ALIGNMENT PLUG: PLUG SHALL PASS COMPLETELY THROUGH THE DRILLED HOLES IN THE HEAD WITHOUT DEFLECTION.

9. TENSILE STRENGTH: BASED ON MINIMUM ULTIMATE TENSILE OF 180,000 PSI FOR SIZES #4 (.112) THRU #10 (.190) AND 170,000 PSI FOR SIZES OVER #10 (.190). LOAD POUNDS CALCULATED BY THE STRESS AREAS INDICATED IN FED-STD-H28/2.

10. FINISH CODE:

    A - ALUMINUM COATING
    U - NO PROTECTIVE COATING OR SURFACE TREATMENT
    NO CODE - CADMIUM PLATE (NOTE: MS24677C PART NUMBERS IN GOVERNMENT INVENTORY WITH ADDED "C" SHALL BE USED UNTIL STOCK IS DEPLETED.)

11. PART NUMBER: THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER TAKEN FROM TABLE I, FOLLOWED BY A FINISH CODE LETTER, IF APPLICABLE.



MS24677-1A INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL; .112-40 UNC-3A; LENGTH .250; ALUMINUM COATING.

MS24677-1 INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL; .112-40 UNC-3A; LENGTH .250; CADMIUM PLATE.

MS24677-1U INDICATES - SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, .112-40 UNC-3A, LENGTH .250; NO PROTECTIVE COATING OR SURFACE TREATMENT.

12. SOURCE MARKING IDENTIFICATION: SCREWS WITH NOMINAL SIZES .190 AND LARGER SHALL BE PERMANENTLY MARKED TO IDENTIFY THE SOURCE ACCEPTING RESPONSIBILITY FOR THE SCREWS MEETING THE REQUIREMENTS SPECIFIED HEREIN. THE MARKING SHALL BE A SOURCE IDENTIFYING SYMBOL FOR A MANUFACTURER IN ACCORDANCE WITH GOVERNMENT REGULATIONS OR A PRIVATE LABEL DISTRIBUTOR'S SYMBOL AS APPLICABLE.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.

2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE.

3. REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN.

PREPARING ACTIVITY: ARMY-AR CUSTODIANS: ARMY- . NAVY- OS AIR FORCE- 99 DLA- REVIEW: MC,MI,SH,IS,11 USER: AS,AT,ME PROJECT NUMBER: 5305-1916	MILITARY SPECIFICATION SHEET TITLE SCREW, CAP, SOCKET HEAD, HEXAGON, DRILLED ALLOY STEEL, (UNC-3A)	SPECIFICATION SHEET NUMBER MS24677D 2 JAN 92 SUPERSEDING MS24677C 5 MARCH 1986 AND MS24676 AMSC- N/A FSC - 5305
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